

1 Solution — GIAM 6.3

Problem 5

1.1 Problem

Let $\mathbb{Z}^* = \mathbb{Z} \setminus \{0\}$. Define a relation Q on $\mathbb{Z} \times \mathbb{Z}^*$ by

$$(a, b) Q (c, d) \iff ad = bc.$$

Show that Q is an equivalence relation.

1.2 Setup

The set $\mathbb{Z} \times \mathbb{Z}^*$ consists of pairs (a, b) with $a \in \mathbb{Z}$ and $b \neq 0$. Read each pair as a placeholder for the fraction a/b (with division by zero forbidden); the cross-multiplication test $ad = bc$ is the familiar rule " $a/b = c/d$ iff $ad = bc$." Showing that Q is an equivalence relation amounts to showing that this cross-multiplication test behaves like equality on the (yet-to-be-constructed) rationals.

We rely only on the basic algebraic properties of $\mathbb{Z}$:

- Multiplication is **commutative** and **associative**.
- $\mathbb{Z}$ has **no zero divisors**: if $xy = 0$ with $x, y \in \mathbb{Z}$, then $x = 0$ or $y = 0$.
Equivalently, **cancellation** holds: $xz = yz$ and $z \neq 0$ together imply $x = y$.

1.3 Reflexive

For any $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$,

$$ab = ba \quad (\text{commutativity}),$$

so $(a, b) Q (a, b)$.

1.4 Symmetric

Suppose $(a, b) Q (c, d)$, i.e. $ad = bc$. Then

$$cb = bc = ad = da,$$

so $cb = da$, which is $(c, d) Q (a, b)$. (Cross-multiplication is symmetric on its face — a single equality, read both ways.)

1.5 Transitive

Suppose $(a, b) Q (c, d)$ and $(c, d) Q (e, f)$, so

$$ad = bc \quad \text{and} \quad cf = de.$$

We aim to prove $af = be$.

Multiply the first equation by f and the second by b :

$$adf = bcf, \quad bcf = bde.$$

Chaining gives $adf = bde$, i.e.,

$$(af)d = (be)d.$$

Now $d \in \mathbb{Z}^*$, so $d \neq 0$. Cancellation in $\mathbb{Z}$ (no zero divisors) then yields

$$af = be,$$

which is exactly $(a, b) Q (e, f)$.

Therefore Q is transitive. Combined with reflexivity and symmetry, Q is an equivalence relation. ■

The hypothesis $d \neq 0$ is essential: cancelling d in the last step requires d to be non-zero. This is why the second coordinate is restricted to $\mathbb{Z}^*$. Were that restriction lifted, the chain

$$(1, 1) Q (0, 0) Q (1, 2)$$

would be valid step-by-step ($1 \cdot 0 = 1 \cdot 0$ and $0 \cdot 2 = 0 \cdot 1$), yet $(1, 1) Q (1, 2)$ would *not* hold (since $1 \cdot 2 \neq 1 \cdot 1$) — and transitivity would fail.

1.6 Remark — The Quotient Is $\mathbb{Q}$

The quotient set

$$\mathbb{Q} := (\mathbb{Z} \times \mathbb{Z}^*) / Q$$

is the standard construction of the **rational numbers** from the integers. The equivalence class of (a, b) is the fraction a/b , and the cross-multiplication test $ad = bc$ is precisely the rule “two fractions are equal iff their cross-products agree.” For instance, the class of $(1, 2)$ is

$$[(1, 2)]_Q = \{(c, 2c) : c \in \mathbb{Z}^*\} = \{\dots, (-2, -4), (-1, -2), (1, 2), (2, 4), (3, 6), \dots\}$$

— every “ $k/2k$ with $k \neq 0$,” all the pairs that represent the rational number $\frac{1}{2}$.

The construction generalizes: replace $\mathbb{Z}$ by any **integral domain** R (a commutative ring with no zero divisors) and $\mathbb{Z}^*$ by $R \setminus \{0\}$. The resulting quotient is the **field of fractions** $\text{Frac}(R)$. So this exercise is the entry point to a foundational construction in algebra: every integral domain sits inside a field, built from itself by exactly this equivalence.